

Jet-image tagging and anomaly detection: a lightweight CNN-based approach

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Abstract—Identifying the physical origin of hadronic jets is an important task in High Energy Physics. While representing jets as two-dimensional images has enabled the highly accurate application of deep Convolutional Neural Networks for classification, state-of-the-art models often suffer from high computational complexity. Furthermore, their strict reliance on supervised learning makes them insensitive to anomalous particle signatures that are not physically modelled yet. To address these challenges, this report presents the *miniEnsemble*, a lightweight CNN architecture for jet image analysis. Evaluated on the JetNet dataset, our model achieves competitive multi-class classification accuracy against established, heavier baselines. Extending its applicability, we adapt the *miniEnsemble* into an unsupervised Convolutional Autoencoder. By training exclusively on standard Quantum Chromodynamics (QCD) background jets, we exploit the reconstruction loss as an anomaly score to successfully isolate anomalous features in jets generated by heavy particles. Overall our analysis demonstrates that a computationally efficient, spatially-aware architecture can effectively bridge the gap between supervised jet tagging and model-independent anomaly detection for new physics searches.

Index Terms—High Energy Physics, Jet Image Tagging, Anomaly Detection, Deep Learning, Convolutional Neural Networks, Autoencoders.

I. INTRODUCTION

In understanding the properties of particles produced in high-energy heavy-ion collisions at the Large Hadron Collider (LHC) experiments, jets play a critical role. Jets are narrow cones of collimated hadron sprays, generated when energetic quarks and gluons undergo fragmentation and hadronization [1], and pose a challenge for identification due to their complex structure and difficult reconstruction. Achieving a reliable classification of jet remnants based on the particle that originated them is of fundamental importance for both achieving high precision measurements and for potentially discovering new physics beyond the Standard Model (SM).

Recent advancements in deep learning applications for particle physics have increasingly improved jet tagging capabilities, with complex architectures operating directly on full constituent particle information, such as Particle Transformers [2], ParticleNet [3] and Lorentz Invariant Networks [4]. An alternative method to process reconstructed jets data is to visualize their content as a compressed image [5]. Jet images act as two-dimensional representations of the relative transverse momentum (p_T) distribution of constituent particles

mapped onto the relative pseudorapidity - azimuthal angle ($\eta^{\text{rel}}, \phi^{\text{rel}}$) plane, centered on the jet's geometric centre. In this domain, Convolutional Neural Networks (CNNs) and Vision Transformers have obtained remarkable success in discerning different jets by extracting features of their geometric distribution in space [6], [7].

In this work, we primarily focus on reducing the high computational complexity typically associated with these deep architectures. Building upon the work of Bassa et al. [7], we propose a smaller-scale model, the *miniEnsemble*, which achieves competitive multi-class classification accuracy on the JetNet dataset [8], [9], while significantly lowering the number of trainable weights. Furthermore, we extend the applicability of our lightweight architecture to address a critical unsupervised task in the field: anomaly detection. Identifying anomalous jet substructures is essential for high-level trigger systems to select relevant events and for the model-independent search of new physics beyond the SM [10]. By appending a decoder to the *miniEnsemble*, we repurpose it into a Convolutional Autoencoder (CAE). This allows us to leverage its reconstruction loss, an established technique in literature for isolating out-of-distribution samples [11], [12], to detect anomalous signatures without prior knowledge of the underlying signal. The main contributions of our work are the following.

- We reproduce the results obtained by Bassa et al. for multi-class jet image tagging with the Ensemble Model using a different learning framework, with a smaller dataset and without using pre-trained weights.
- We propose a smaller-scale model, the *miniEnsemble*, with which we obtain comparable accuracy in the classification task while reducing trainable parameters by 96%.
- We demonstrate the increased applicability of our model, exploiting it to tackle a different, unsupervised task: the detection of anomalously-shaped jets generated by particles not present in the training dataset. For this task we attached a decoder network to the *miniEnsemble* and used it as a CAE tailored for the sparse and localized nature of energy deposits in jet images.

The rest of this report is structured as follows. In Section II we review the state of the art in jet tagging and anomaly detection. In Sections III and IV we respectively describe the learning architectures and the processing of jet data. The proposed learning framework is detailed in Section V, while its performance evaluation is carried out in Section VI. Finally, conclusions are drawn in Section VII.

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II. RELATED WORK

The representation of collider data as jet images, proposed by Cogan et al. [13] and comprehensively reviewed by Kagan et al. [5], provided an efficient alternative to processing raw kinematic point clouds. Following this path, several works successfully applied CNNs to discriminate quark and gluon jets or classify boosted heavy resonances [6], [14]. Recently, Bassa et al. [7] proposed an Ensemble Model that combines parallel CNN architectures typically used for standard real-life images classification (e.g., ResNet and Inception) to achieve state-of-the-art accuracy. These deep frameworks demonstrate good potential in extracting relevant features from calorimeter data, however they rely on architectures containing tens of millions of parameters which may not be necessary to analyse highly sparse jet images.

To enable model-independent searches, the HEP community has increasingly explored unsupervised anomaly detection [10]. Early applications by Farina et al. [11] and Heimes et al. [12] proposed the use of autoencoders trained on background QCD jets, exploiting the reconstruction loss as a metric to isolate anomalous signals. These studies successfully proved the viability of reconstruction-based tagging.

Our project positions itself within this context by primarily aiming to optimize the computational efficiency of image-based jet taggers. We start from the CNN-based classifier proposed by Bassa et al. [7] and introduce a lightweight architecture (the *miniEnsemble*) designed to reduce the computational burden while maintaining competitive classification performance. Furthermore, as an exploratory step to test the versatility of our architecture, we adapt the *miniEnsemble* into a Convolutional Autoencoder framework. By evaluating this repurposed model on an unsupervised anomaly detection task, we aim to investigate whether such architecture can be effectively utilized as a tool for identifying out-of-distribution signatures in jet images.

III. PROCESSING PIPELINE

A. Ensemble Models for Classification

As a baseline for our classification pipeline we implemented the architecture proposed by Bassa et al. [7], which processes input jet images through a parallel ensemble of an InceptionV3 and a ResNet50 module. The outputs of both structures are directly concatenated and passed through a small MLP for the final classification.

To address computational complexity of this baseline, we designed an optimized architecture, the *miniEnsemble*, schematically represented in Fig. 1. Our optimization was guided by the following principles, established from the findings in the original Inception and ResNet works [15], [16]:

- We maintained a structural equilibrium between the *miniInception* and *miniResNet* modules, avoiding restrictive bottlenecks that could discard essential information.
- The reduced modules preserve their defining characteristics: multi-scale convolutional operations for the Inception blocks, and residual skip-connections for the ResNet blocks.

- We followed the standard design rule of increasing the number of channel filters whenever the spatial dimensions of the feature maps are downsampled.

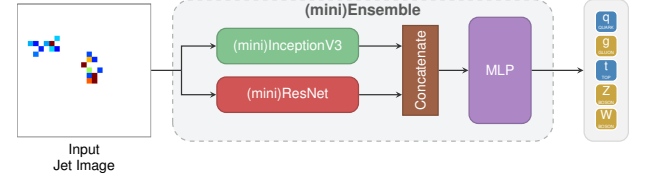


Fig. 1: Schematic representation of the *miniEnsemble* architecture for multi-class jet tagging.

B. Autoencoder for Anomaly Detection

For the unsupervised anomaly detection task, we repurposed our optimized architecture into a CAE, as depicted in Fig. 2.

The encoder module directly inherits the *miniEnsemble* architecture used for the classification task. The final classification layer is removed and replaced with a dense layer composed of a reduced number of neurons, which acts as the latent space. This yields a latent vector $z \in \mathbb{R}^L$, where the bottleneck dimension L is deliberately kept small.

The decoder module consists of a straightforward sequence of upsampling convolutional layers aimed at reconstructing the original input image from z . Again, this network is mainly designed to maintain a standard structural equilibrium between the spatial dimensions of the feature maps and the number of channel filters in each layer.

The CAE is trained exclusively on standard background jets (i.e., quarks and gluons). During inference, images from other classes are fed to the network: these images inherently produce a larger reconstruction error, which is then used as the anomaly score to tag them.

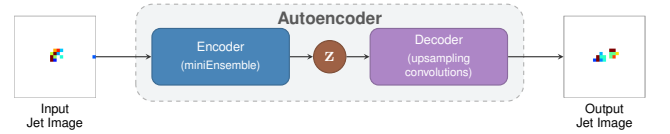


Fig. 2: Autoencoder architecture schematic representation.

IV. SIGNALS AND FEATURES

To train the models proposed in this work we used the JetNet dataset, introduced by Kansal et al. [9] for machine learning applications in High Energy Physics. It contains point-cloud representations of particle jets produced in proton-proton (pp) collisions at a center-of-mass energy of $\sqrt{s} = 13$ TeV, clustered using the anti- k_t algorithm with a radius parameter $R = 0.4$ [17].

For each jet, the information available on its particle constituents is purely kinematical. Three values are stored for every particle: the relative transverse momentum p_T^{rel} , the relative pseudorapidity η^{rel} , and the relative azimuthal angle ϕ^{rel} evaluated with respect to the jet axis. Jets are categorized into five distinct classes based on their originating source:

gluons (g -jets), light quarks (q -jets), top quarks (t -jets), W bosons (W -jets), and Z bosons (Z -jets).

In our setup, a jet-image is defined as a two-dimensional binned histogram in the $(\eta^{\text{rel}}, \phi^{\text{rel}})$ plane, where the pixel intensity corresponds to the accumulated p_T^{rel} of the 30 highest- p_T constituent particles.

To prevent high image sparsity, we restricted the parameter space from its original ranges ($\eta^{\text{rel}} \in [-1.6, 1.0]$ and $\phi^{\text{rel}} \in [-0.5, 0.5]$) to a narrower region: $\eta^{\text{rel}} \in [-0.4, 0.4]$ and $\phi^{\text{rel}} \in [-0.4, 0.4]$. Selecting wider bins would reduce the spatial resolution of particle distribution, whereas smaller bins would make the image excessively sparse. Considering this trade-off, we selected a resolution of 128×128 pixels. After applying these spatial filters, the total available dataset consists of 880,000 images, equally balanced across the five classes.

Although data augmentation is standard practice in computer vision, applying conventional spatial or intensity transformations directly at the jet-image level introduces physical inconsistencies that would have a detrimental impact on the final accuracy. For instance, continuous rotations in the $(\eta^{\text{rel}}, \phi^{\text{rel}})$ plane alter the underlying physics, as this space is not Euclidean isotropic and Lorentz symmetry applies differently along beam and azimuthal directions. For our experiments, we resorted to applying pepper noise (setting random pixels to zero intensity, thus effectively removing particles) to the images as the only augmentation strategy. This simulates the possible presence of dead channels in a detector, even though it strictly violates four-momentum conservation by removing energy from the system.

To maintain manageable training times and accommodate computational resource constraints, the data was subsampled and then partitioned into an 80% / 10% / 10% split for training, validation, and testing, respectively. This split was performed in a stratified manner to preserve the class proportions across all subsets. Depending on the task, the samples were then collected as follows:

- **Classification:** we randomly subsampled a total of 150,000 images, ensuring class balance by selecting 30,000 samples per class.
- **Anomaly Detection:** since the autoencoder must learn solely the standard background distribution, we initially sampled images exclusively from the chosen background classes (light quarks $[q]$ and gluons $[g]$, representing a physical QCD background), using the same splitting proportions. To construct the final evaluation set, we took the resulting background test split and shuffled in an equal number of anomalous jets, subsampled evenly from the three remaining signal classes ($[W]$, $[Z]$, and $[t]$, i.e. the heavy particles available in our set). In this way we built two datasets: a smaller one with a total of 30,000 images and a bigger one with 150,000 of them.

V. LEARNING FRAMEWORK

A. Classification task: *miniEnsemble* training

As introduced in Section III, our objective is to correctly classify jets into five categories (g, q, t, W, Z). We engineered

the *miniEnsemble* architecture by systematically scaling down the standard ResNet and Inception modules to reduce by 95% the amount of trainable weights with respect to the original Ensemble model by Bassa et al [7]. We then trained both the baseline and the *miniEnsemble* independently, applying the same procedure described at the end of this subsection.

The *miniResNet* branch (Fig. 3) begins with a stem block comprising a 7×7 convolution with a stride of 2 (64 filters), followed by Batch Normalization, a ReLU activation, and a 3×3 Max Pooling layer. The core feature extraction is performed by a sequence of three residual blocks with channel dimensions of 64, 128, and 256, respectively. Strided convolutions within the second and third blocks progressively halve the spatial dimensions. To prevent overfitting, a mild dropout layer ($p = 0.2$) is inserted after each residual block. An average pooling layer finally collapses the spatial dimensions, yielding a 256-dimensional feature vector.

The *miniInception* branch (Fig. 3) is designed to capture multi-scale spatial correlations. Its stem applies a sequence of three convolutional layers with 3×3 kernels up to 64 filters and a max pooling layer, reducing the size of feature maps to 32×32 . The network then continues with two *Inception-A* blocks, which utilize parallel 1×1 , 3×3 , and 5×5 convolutional branches to process the spatial information at different receptive fields, reaching 128 channels. A reduction block subsequently downsamples the resolution to 16×16 while expanding the depth to 256 channels. Finally, an *Inception-B* block applies parallel convolutions with 1×7 and 7×1 kernels, expanding the feature space to 512 channels. Following Adaptive Average Pooling and a Dropout layer ($p = 0.2$), this branch outputs a 512-dimensional vector.

The *miniEnsemble* model (Fig. 1) merges these representations by concatenating the *miniResNet* and *miniInception* output feature vectors, yielding a map of shape $[B, 768]$. This unified representation is passed through a MLP head consisting of a 128-neuron dense layer, a ReLU activation, a Dropout layer, and a final linear layer mapping to the 5 target classes.

We applied a softmax activation to the final layer, yielding a predicted probability distribution $\hat{y}_i \in [0, 1]^C$ across the $C = 5$ target classes for each input image. The network was then iteratively updated using the Adam optimizer to minimize the Categorical Cross-Entropy loss, defined as:

$$\mathcal{L}_{\text{CCE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}), \quad (1)$$

where N is the batch size and $y_i \in \{0, 1\}^C$ denote the one-hot encoded true label of each sample. The training procedure was divided into two distinct phases: an independent pre-training of the branch backbones, followed by a fine-tuning of the entire ensemble. Before proceeding with each phase, hyperparameter tuning was performed using the *optuna* framework, in order to select the best combination of learning rate, batch size and weight decay (L2 regularization) constants.

Phase 1: Independent Pre-training. Initially, the *miniIn-*

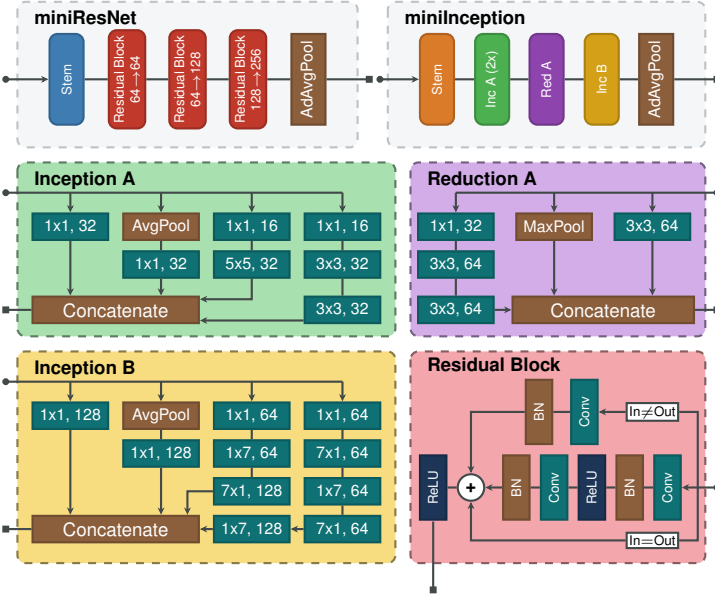


Fig. 3: Schematic representations of the miniResNet and miniInception architectures, along with their building blocks.

ception and *miniResNet* modules were trained independently, adding an opportune dense final layer with 5 output neurons to their structure. For the *miniInception*, the optimal setup utilized a learning rate of 1×10^{-4} , a weight decay of 1×10^{-4} , and a batch size of 32. For the *miniResNet*, we employed a learning rate of 1×10^{-4} , a weight decay of 1×10^{-5} , and a batch size of 64. Both models were trained for a maximum of 30 training plus validation epochs, implementing an early stopping strategy with a patience of 5 epochs if the validation loss did not improve. The optimized weights from this phase were saved to initialize the backbones of the full *miniEnsemble*.

Phase 2: Ensemble Fine-tuning. Once the backbones were loaded with their pre-trained weights, the full *miniEnsemble* was trained for 30 additional epochs. To prevent the randomly initialized weights of the MLP head from destructing the optimized feature extractors, we implemented a progressive unfreezing strategy. During a 3-epoch warm-up phase, the backbone weights were frozen, allowing only the MLP head to stabilize. Subsequently, the backbones were released, enabling an end-to-end fine-tuning. In this phase, through *optuna* tuning, we set differential learning rates: 1×10^{-4} for the MLP head and a lower 1×10^{-6} for the backbones. The MLP dropout was fixed to $p = 0.38$, with an overall model weight decay of 1×10^{-4} and a batch size of 64. The same early stopping protocol was maintained.

B. Anomaly detection

The architecture we designed for anomaly detection is a CAE, schematically shown in Fig. 2, with the encoder module directly adapted from the *miniEnsemble* we employed during classification. The 768-dimensional concatenated output of *miniResNet* and *miniInception* is directly passed to a single fully-connected linear layer, that maps its features to an

L -dimensional vector in the latent space. For the decoder module we opted for a simpler architecture based on upsampling convolutions. Concretely, the latent space vector is first passed through a two-layer dense MLP, with 256 and 4096 features respectively, after which the output is unflattened into a $256 \times 4 \times 4$ tensor. Then, a series of 5 consecutive transposed convolutions upsamples the tensor into a single 128×128 image. Each convolution is performed with filters of size $K = 4$, stride $S = 2$ and padding $P = 1$; after each block a Batch Normalization layer was used along with a ReLU activation function, except for the last layer, where an identity activation is adopted for regression. We were careful to maintain an overall symmetry in the number of weights of the encoder and decoder modules, respectively around 1.9×10^6 and 1.8×10^6 . However, the decoder structure is indeed simpler in the information flow compared to encoder one: this asymmetry between the two modules could help in both preventing the autoencoder from learning the identity map and shifting the learning effort slightly towards the encoder, forcing it to learn a more robust latent space representation.

As in any standard autoencoder implementation, the loss function is computed as the MSE for image reconstruction, averaged over all the pixels in the image. In particular, given a batch with N images of $S = 128 \times 128 = 16384$ pixels each, the loss function we minimized is

$$\mathcal{L}_{AE} = \frac{1}{NS} \sum_{n=1}^N \sum_{i=1}^S \left(x_i^{(n)} - \hat{x}_i^{(n)} \right)^2. \quad (2)$$

Also in this implementation, we chose to use Adam as the optimizer.

As anticipated in Section III, reproducing a faithful representation of the input jet image is not the ultimate purpose of this network, but rather accurately detecting anomalous signals over the background. The anomaly score we defined for this scope is the MSE in the image reconstruction, analogous to the loss function in Eq. 2.

Training Strategy and Architecture Tuning. The training procedure mirrored the methodology of the classification task. Initial hyperparameter tuning yielded an optimal learning rate and weight decay of 1×10^{-4} , a batch size of 64, and a bottleneck dimension of $L = 16$. After this tuning phase, we trained the network twice: first on the small dataset, then on the larger one, as described in Section IV.

During the tuning phase we also evaluated alternative architectural formulations and regularization strategies with respect to the one described up to now:

- A Variational AE (VAE). We argued that a random latent variable could better grasp the intrinsic probabilistic nature of the experimental input data. In this variant, the model optimizes a composite loss function combining $\mathcal{L}_{AE}$ with the Kullback-Leibler divergence term.
- A Sparse AE (SAE). Given the sparsity of jet images, we attempted to force a sparse internal representation by adding a L1 regularization term to the loss: $\mathcal{L}_{SAE} = \mathcal{L}_{AE} + \lambda \sum_{j=1}^L |z_j|$, with the optimal penalty pa-

Model	Train Acc.	Val. Acc.	Test Acc.	Avg. AUC	Time (min)	N° par.
<i>miniResNet</i>	0.7396	0.7249	0.7256	0.923	1372	1.2×10^6
<i>miniInception</i>	0.7588	0.7427	0.7450	0.928	1430	0.6×10^6
<i>miniEnsemble</i>	0.7801	0.7505	0.7494	0.932	1553*	1.9×10^6
Ensemble	0.7980	0.7425	0.7456	0.929	575*	47.4×10^6
ResNet50 [7]	0.7383	0.7149	0.7248	0.92	2009	23.5×10^6
Inceptionv3 [7]	0.7525	0.7217	0.7369	0.923	2013	21.8×10^6
Ensemble [7]	0.7868	0.7298	0.7508	0.935	2052	47.4×10^6

TABLE 1: Performance metrics for multi-class classification. Comparison of our models with the results obtained by Bassa et al. [7]. (*) Training times for ensemble models only account for the fine-tuning phase.

parameter found to be $\lambda = 0.005$.

- An AE with AdaGrad optimizer (AGAE). As suggested by Kasieczka et al. [10], the AdaGrad optimizer might be better suited for the reconstruction of sparse images. In this case we started with a higher learning rate of 0.01.

VI. RESULTS

A. Multi-class jet classification

To quantitatively evaluate the performance of our models, we use three main metrics: overall accuracies, the Receiver Operating Characteristic (ROC) curves for each class and the average Area Under these Curves (AUC). We also plotted normalized confusion matrices onto the test set to evaluate separately the tagging performance onto each class.

Quantitative results are summarized in Table 1, where the performance of our light models are compared against individual substructures and the baselines of [7]. First, we observe that Ensemble architectures perform better than individual branches in both scenarios, standard and lightweight. Our *miniEnsemble* achieves a test accuracy of 74.94% and an average AUC of 0.932, closely matching the performance of the heavier baseline Ensemble (75.08% test accuracy and 0.935 AUC) while reducing by 96% the number of trainable weights.

Looking at the *miniEnsemble* test confusion matrix in Fig. 4b, the network exhibits a good discrimination power across different topological jet signatures, as it cleanly separates multi-prong heavy particle (t, W, Z) jets from single-core QCD (q, g) ones. This aligns with the underlying particle decays: top quarks decay through the $t \rightarrow Wb \rightarrow q\bar{q}'b$

chain forming 3 sub-jets, W and Z bosons undergo two-body hadronic decays ($W \rightarrow q\bar{q}'$ and $Z \rightarrow q\bar{q}$) forming 2 sub-jets, and QCD background jets originate from single partons (q/g, 1 sub-jet only). As expected, misidentifications primarily occur within topologically equivalent categories: between quark and gluon jets due to overlapping shower profiles with a single energy deposit, and between 2-prong W and Z jets because of their matching multiplicity and comparable masses.

B. Anomaly Detection

To evaluate model performance we again used ROC curves and AUC as our main metrics. To understand the effectiveness of the network in separating the background from the signals, we plotted the anomaly scores histogram distribution and the 2D latent space obtained through PCA for the baseline CAE (Fig. 5). We summarize the results in Tab. 2 below.

Model	AUC	Val. Loss	Model	AUC	Val Loss
AE - 30k	0.830	0.151	AGAE	0.794	0.621
AE - 150k	0.771	0.094	SAE	0.829	0.238*
VAE	0.811	0.189*	* Loss has an extra term		

TABLE 2: Validation loss and AUC performance summary for different Autoencoder models.

Remarkably, the CAE trained on the smaller dataset outperformed the one trained on the bigger dataset. This is particularly counter-intuitive, since one would expect a network to perform better when trained on more data. We believe that showing the model a bigger variety of samples exposed it to more "extreme" jet image configurations, which could potentially have many more features in common with the anomalous ones. Looking at Fig. 5c and 5b, we can see that this is already true for the AE - 30k model, being the histograms partially overlapped and the latent space not sharply clustered. On the other hand, considering all the necessary measures we employed, we are prone to exclude that the network has just started to learn the identity map.

Regarding the experimental variations of the baseline CAE (all trained on the smaller dataset), we note that these solutions did not provide any true improvement in performance, sometimes even worsening it. The VAE implementation failed

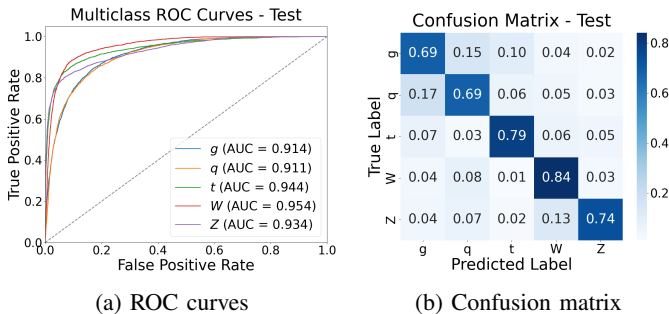


Fig. 4: *miniEnsemble* test performances.

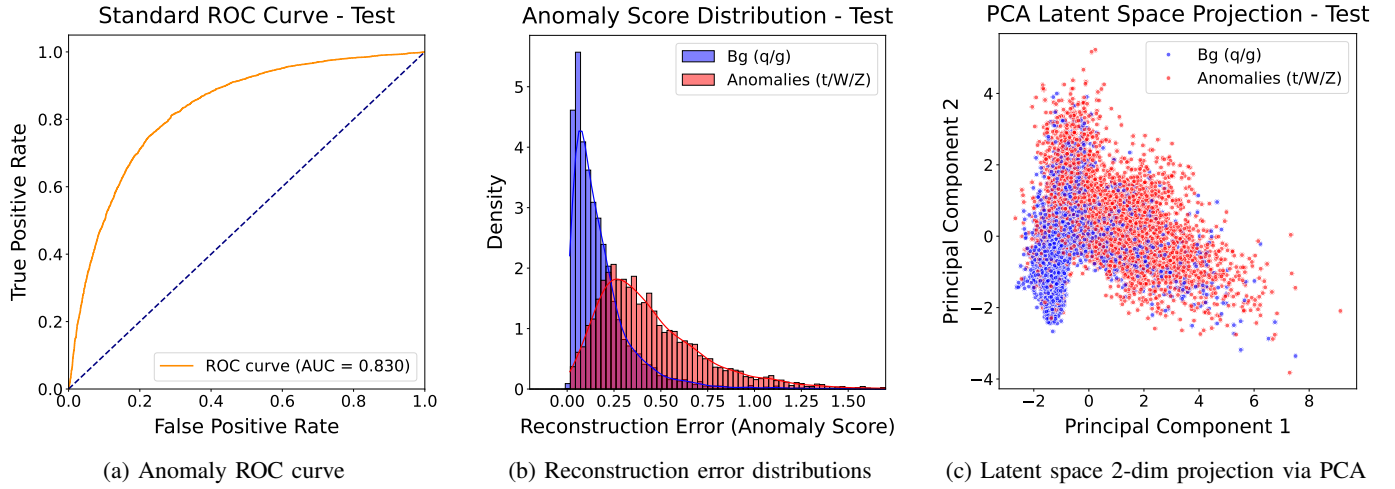


Fig. 5: Performance plots of the AE model trained on a dataset of 30,000 background samples.

to meet our expectations: the greater flexibility provided by a probabilistic latent space did not allow the model to better grasp the characteristic features of the background signals. On the contrary, it seems that the versatility of this model, compared to the baseline AE, made it more prone to adapt to anomalous jets. The AdaGrad optimizer, in spite of reconstructed images that were visibly sparser compared to the ones obtained with Adam, showed significantly poorer performance during training. Finally, forcing the latent space to be sparse only produced results that are comparable with the baseline implementation.

VII. CONCLUDING REMARKS

In this work we implemented *miniEnsemble*, a CNN-based architecture for jet images analysis. The model was used in multi-class image tagging, obtaining state-of-the-art accuracies while reducing the number of trainable weights by 96% in comparison with literature baselines.

Having fast-triggering applications during particle-data acquisitions in mind, we easily adapted the model for self-supervised anomaly detection, introducing a Convolutional Autoencoder architecture. The obtained results suggest that, despite good overall performances, we extrapolated most of the information available and further improvements are difficultly achievable without drastic changes to the working paradigm.

The greatest limitation of this framework is indeed represented by the intrinsic sparsity of jet images, making the relevant features particularly fragmented and difficult to access in their integrity. In this regard, above mentioned neural networks based on point clouds or graph analysis are capable of a more efficient information extraction, while having to deal, on the other hand, with much more complex architectures. Significant challenges were also encountered in the miniaturization process of our model, since we had to be careful not to destroy the strong feature-extraction capabilities of the baselines in the effort of reducing their complexity.

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